

Scalable Multi-User Gaze Estimation for Co-located Environments Using Commodity Webcams

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Abstract—Multi-user gaze interaction traditionally relies on dedicated eye trackers per participant, limiting scalability and practical deployment in shared interactive environments. In this work, we investigate the feasibility of an architecture for real-time multi-user gaze estimation using commodity hardware. While recent advances in appearance-based gaze estimation enable gaze prediction using RGB cameras, it remains unclear whether such methods can support multi-user behavioral analysis in realistic environments with low-cost sensors. Rather than proposing a new gaze estimator, we integrate existing gaze estimation and gaze-to-screen mapping methods into a unified real-time pipeline and analyze their effectiveness for shared interaction scenarios. For this purpose, we introduce a multi-user gaze interaction dataset that captures joint and non-joint attention behaviors on a shared interface, and we perform an end-to-end evaluation of the unified gaze pipeline. While spatial precision remains lower than that of dedicated eye trackers, our results demonstrate that coarse gaze localization is sufficient for detecting collaborative attention patterns. Our findings characterize key accuracy-scalability tradeoffs and suggest the practicality of commodity hardware-based multi-user gaze tracking without per-user instrumentation.

Index Terms—Multi-user eye-tracking, Commodity hardware, Gaze estimation

I. INTRODUCTION

The ocular system provides informative cues for understanding human behavior and functions as a mode of interaction across a wide range of activities [1]. Eye-tracking techniques have broad applications in human-computer interaction [2], psychology [3], and behavioral sciences [1]. However, building accurate mappings from eye appearance to gaze position requires complex functions [4], [5], and traditional eye trackers address this through specialized hardware with IR illumination, limiting their practical deployment due to cost and operational complexity [6], [7].

As a cost-effective alternative, commodity web cameras have gained increasing attention for gaze estimation [8], [9]. By leveraging advances in computer vision [5], [10], gaze direction can be estimated from facial landmarks, eye appearance, and head pose, significantly reducing cost and simplifying deployment for scalable, real-world applications.

Although single-user eye trackers perform well in isolation, they cannot simultaneously capture gaze from multiple users [11]. The trivial solution of dedicating one tracker per user [12]–[14] introduces interference, higher cost, and deployment constraints. Recent advances in appearance-based gaze estimation [10] and large-scale datasets [15], [16] suggest the

potential for scalable shared eye-trackers, yet few studies have demonstrated their utility in multi-user environments [1], [11].

Multi-user eye tracking aggregates gaze measures across two or more participants engaged in collaborative activity, enabling gaze awareness that improves collaborative outcomes [12], [13] and supports quantitative analysis of joint attention [14], [17]. Despite this broad utility, limited research exists on system design and evaluation for such settings. Therefore, we propose Multi-Eyes, a framework for multi-user eye-tracking that encompasses the capability to perform eye-tracking, generate corresponding measures, and provide visualizations for the generated measures.

This paper makes the following contributions:

- 1) A multi-user gaze estimation architecture for dyadic co-located interactions
- 2) A characterization of accuracy-scalability tradeoffs under varying calibration strategies and mapping techniques
- 3) A multi-user region-level gaze dataset capturing joint and non-joint attention behaviors
- 4) An empirical evaluation of multi-user gaze estimation performance under personalized and non-personalized calibration settings

II. RELATED WORK

The existing multi-user eye-tracking setups can be broadly classified as time-sharing and space-sharing approaches [4]. Time-sharing approaches aggregate eye-tracking data collected across non-overlapping time windows to simulate a multi-user environment [18], [19], but fail to capture collaborative behaviors due to the lack of simultaneity. In contrast, space-sharing approaches capture gaze data concurrently, enabling the study of collaborative interaction [4].

The main challenge in space-sharing setups is the lack of hardware or software capable of tracking more than one participant simultaneously [11], [14]. The most common solution is to assign a dedicated eye tracker to each participant [12], [14], [17]. DisETrac [14] exemplifies this approach by aggregating eye-tracking data over a public network using the MQTT protocol [20], with periodic clock synchronization via NTP [21] to handle clock drift. In addition to nodes with eye trackers, the system also proposes nodes that offer gaze aggregation and analytics in the form of a dashboard. Look-Together [12] demonstrates a similar dedicated setup in co-located environments, showing the potential of gaze-assisted

techniques for collaborative interaction. Pfeuffer et al. [22] demonstrate a co-located setup using multiple eye trackers on a shared touchscreen, highlighting the benefits of gaze-assisted interaction but also the scalability limitations of dedicated hardware. However, these setups do not scale beyond dyadic interactions due to cost and movement restrictions [11].

Shared eye-tracking systems using standard RGB cameras offer a more scalable alternative [19], [23]. One approach aggregates individual gaze directions across a crowd to localize a shared target [19], though performance degrades in less crowded settings. GazeOnce [23] addresses this with a single-stage multi-user gaze estimation framework, but offers limited diagnosability for identifying system failures. However, it estimates gaze angles rather than on-screen gaze positions, limiting its applicability for interaction analysis on shared displays. Camera-based approaches using digital twins and synthetic environments have also been proposed [24], [25], though neither evaluates real-time performance. Additionally, most prior work focuses either on distributed environments or dyadic co-located setups, leaving a gap in scalable, real-time shared-device solutions for collaborative interaction [11].

Our work contributes to this space by examining how appearance-based gaze estimation can be engineered into a commodity hardware pipeline that supports simultaneous multi-user attention detection in co-located settings.

III. METHODOLOGY

Our system consists of a multi-user gaze tracking pipeline, including user detection from webcam frames, gaze direction estimation using a pre-trained appearance-based model, and mapping of gaze directions to the screen coordinates for interaction analysis.

A. Multi-user gaze tracking framework

We adopt the multi-user gaze tracking framework proposed by [11], which estimates on-screen gaze positions through a three-stage pipeline: subject detection, gaze estimation, and gaze-to-screen mapping.

In the first step of gaze subject detection, the framework first identifies each user within predefined regions of the camera image. Instead of performing a full scene search, the input frame is partitioned into user-designated areas where participants are expected to appear. Within each region, facial landmarks are detected using the FaceMesh model [26], and a bounding box is generated from the landmark centroid. The cropped facial patches are then forwarded to the gaze estimator. This design eliminates unnecessary operations, such as multi-face search or occlusion reasoning, thereby improving real-time processing capabilities.

1) *Gaze Estimation Model*: For gaze estimation, we used an appearance-based gaze estimation model using a disentangled latent representation. The disentanglement in latent representation follows the intuition of gaze-defining and non-defining features for a given facial image (I). The model first generates a latent representation (z) for each detected facial image using an encoder (E) based on a convolutional

neural network. Instead of directly regressing gaze from the full feature space, the network decomposes the representation into two independent components:

- **Gaze-related latent code** (z_g) — encodes ocular orientation and gaze direction.
- **Appearance-related latent code** (z_a) — captures identity, illumination, and facial structure.

A dedicated gaze decoder (G) predicts the gaze direction ($\hat{\theta}$ for yaw and $\hat{\phi}$ for pitch) in the head-centric coordinate system using only the gaze-related latent variables, preventing leakage of identity-dependent information into the regression task (see Equation 1). Simultaneously, a reconstruction decoder (D) reconstructs the input face using both latent components, enforcing representation consistency and preserving semantic structure. This disentanglement acts as an implicit regularizer that improves generalization across subjects and environmental conditions within the source domain.

$$(\hat{\theta}, \hat{\phi}) = G(z_g) \quad \text{where } z = (z_a, z_g) = E(I) \quad (1)$$

Similar to the work of [27] in unsupervised domain adaptation, where the target domain lacks training data for improving the cross-domain performance, we use an adversarially trained domain classifier (C) attempting to distinguish between source and target domains.

We train the model using the multi-task objectives comprising gaze loss (L_g), image reconstruction loss (L_r), and domain classification loss (L_c). We use λ_r and λ_c as hyperparameters to control the contribution of reconstruction and domain adaptation during training using Equation 2.

$$L = L_g + \lambda_r L_r + \lambda_c L_c \quad (2)$$

We use four models in our analysis, with a standard gaze estimation model serving as the baseline, trained on the source domain. We also use a disentangled model without domain adaptation ($\lambda_r = 0.1$, Source only), a domain-adapted model without adversarial training ($\lambda_r = 0.1, \lambda_c = 0.0$, Source + Target), and a model trained with adversarial training ($\lambda_r = 0.1, \lambda_c = 0.1$, Source + Target). We then use the ETH XGaze [15] as the source domain with Gaze360 [16] as the target domain.

The model demonstrates reasonable generalization across multiple gaze datasets collected under different environments and capture setups. Although performance varies with the target dataset characteristics, the approach is resilient to domain shift and improves after domain adaptation, indicating that the disentangled representation captures transferable gaze features rather than dataset-specific appearance cues. Consequently, the model can operate beyond its original training distribution and remain applicable to real-world scenarios without requiring large amounts of labeled data from the target domain.

2) *Gaze-to-Screen Mapping*: Gaze direction estimates alone are insufficient to determine on-screen gaze locations because the mapping depends on both gaze direction and user position. To map gaze directions to on-screen positions, we

use the position-encoding and mapping scheme proposed by [11]. The encoding system is based on the pinhole camera model and assumes the camera remains fixed relative to the interaction surface. It uses the geometric relationship between the face position in the world coordinates (x_p, y_p, z_p) , its spatial extents $(\Delta x_p, \Delta y_p, \Delta z_p)$ and its projection onto the image plane, which is centered at (u, v) with image dimensions $(\Delta u, \Delta v)$, as determined by the camera’s focal length.

The approach first constructs a binary mask from the facial region in the input image. This mask is then subjected to average pooling by partitioning the image into $n \times n$ non-overlapping regions, yielding a coarse spatial representation $P \in \mathbb{R}^{n \times n}$ that encodes the approximate facial position. Finally, it uses the gaze directions $(\hat{\theta}, \hat{\phi})$, combined with the encoded face position P , to predict the screen coordinates $s = (s_x, s_y)$ via a nonlinear mapping function M (Equation 3).

$$M : (P, g) \rightarrow s \quad (3)$$

Following [11], we set $n = 2$ and employ a multi-layer perceptron (MLP) as the mapping function. In addition, we investigate the feasibility of estimating the mapping function using a nearest-neighbor regressor as an alternative. Consistent with the original framework, we train these models using calibration data.

B. Multi-User Gaze Dataset

We introduce a novel multi-user gaze dataset captured using commodity web cameras during dyadic interactions on a shared interactive surface. The data collection setup records synchronized facial images and corresponding gaze target locations while two participants interact simultaneously in front of the display.

1) *Application Setup*: For the data collection, we developed a custom application that simulates joint and non-joint attention conditions using a set of 25 predefined locations on a screen (Figure 1). To simulate joint attention, the application displays a single green circle, prompting both participants to fixate on the target location (Figure 2). In the non-joint attention simulation (Fig. 3), two differently colored circles appear at separate locations, and each participant fixates on a pre-assigned color. During each data collection session, the application captures images of participants using two webcams and records the locations of the on-screen targets (Figure 4).

Our goal was to develop a practical multi-user eye-tracking system that operates with minimal calibration using commodity hardware. Therefore, we designed the following data collection and evaluation procedure to assess the feasibility of the proposed pipeline under realistic usage conditions.

2) *Participants*: We recruited 11 participants (3 female, 8 male) from the university community, each with normal or corrected-to-normal vision. Participants were briefed on the experiment purpose and task details, and written consent was obtained for facial image and eye-tracking data collection, storage, and distribution, in accordance with institutional policies.

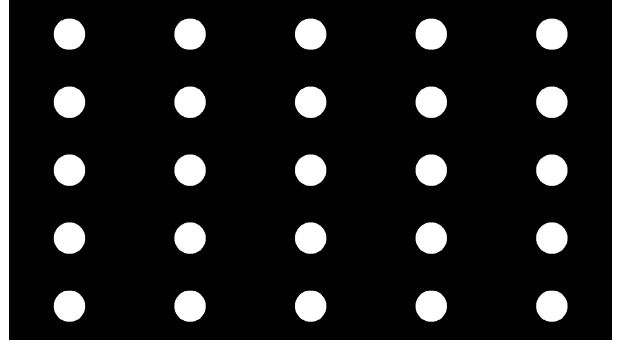


Fig. 1. The grid interface used for multi-user data collection, where the screen is divided into 25 fixed locations, hidden from users.

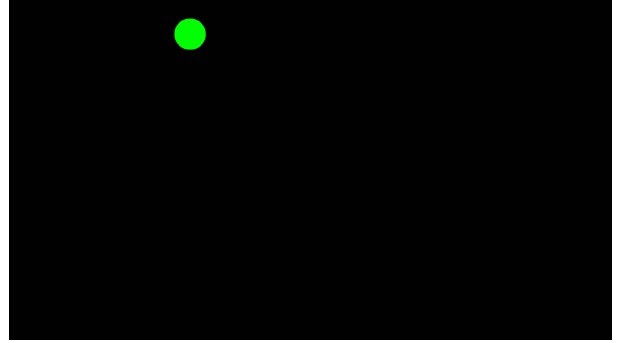


Fig. 2. The interface used to simulate joint attention in data collection by prompting each user to direct gaze to the green circle.

3) *Procedure*: The facilitator positioned participants in the designated capture region and conducted a brief test run before formal data collection began. During each session, the facilitator manually advanced visual stimuli, with a one-second stabilization delay before capturing camera images. Two cameras were positioned at the top and bottom edges of the display (Figure 4) to maximize coverage across multiple viewing angles. Using different participant combinations to form dyads, we conducted 14 sessions of 100 images each, yielding 1400 images in total.

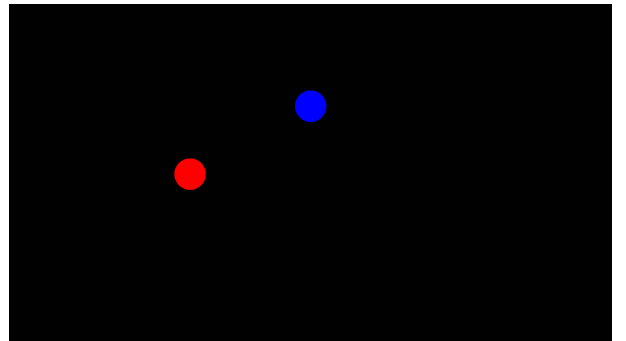


Fig. 3. The interface used to simulate non-joint attention in data collection by prompting each user to direct gaze into the circle with the pre-assigned color.

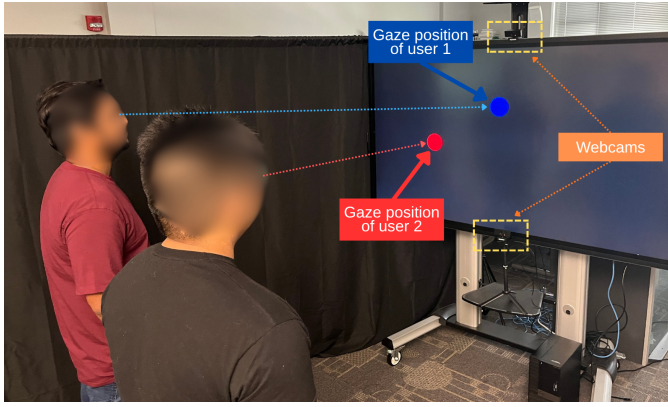


Fig. 4. Data collection setup used for multi-user eye tracking data collection, consisting of two webcams mounted on top and bottom of a shared display, with on-screen gaze targets for each user.

4) *Data Processing*: Captured images were processed through the face detection pipeline, with manual refinement of poorly estimated bounding boxes to ensure accuracy. From the refined bounding boxes, we extracted facial regions, computed gaze directions using the selected gaze estimation models, and derived positional encodings capturing each face’s spatial relationship within the image. Gaze direction estimates and positional encodings were then combined via the mapping function to infer on-screen gaze locations, which were evaluated against ground truth.

IV. RESULTS

We evaluated the performance of the proposed pipeline under two calibration settings. In the first, we simulated a workflow similar to a traditional eye tracker, in which a **personalized** calibration is performed for each session. In the second, we simulated a **non-personalized** eye tracker in which gaze estimation relies on data from an independent session. The second setting is analogous to deploying an eye tracker calibrated for one individual and subsequently used by a different set of individuals.

A. Personalized Calibration

For each user in each session, we conducted 5-point and 9-point calibration procedures using the collected data. The 5-point calibration protocol included the four screen corners and the center point, while the 9-point calibration included central points along each of the four screen edges. We then evaluated the performance of our multi-user eye tracker using non-calibration data samples by measuring gaze estimation error (in millimeters) in the horizontal and vertical directions. We experimented with modelling the function using K-Nearest Neighbors ($K \in \{3, 5\}$) regressor or a Multi-Layer Perceptron (MLP), and report a summary of gaze errors across 14 experiments, for the four models we previously selected (Table I).

The experiments yielded mixed results across various combinations of gaze models and mapping functions (see Table I). Notably, the domain-adapted model consistently performed

the lowest across all configurations. This outcome may be attributed to limitations in the mapping function or the encoding scheme, particularly given the domain-adapted model’s superior performance in other gaze estimation tasks.

However, we observed that the domain-adapted model consistently exhibited the lowest coefficient of variation across all calibration configurations (see Table II). This finding suggests that, despite its overall poorer performance, the domain-adapted model provides more predictable and stable gaze estimates, with lower relative variability than the other models. Such stability could be advantageous in applications where consistent performance is prioritized over absolute accuracy. Further, this indicates the potential of a domain-adapted model to produce more refined estimates through alternative calibration and error-correction mechanisms.

B. Non-Personalized Calibration

In this experiment, we calibrated the eye tracker using the entire dataset from a single session and evaluated gaze estimation across the other 13 sessions. We used the same set of gaze models and experimented with modelling the mapping function using K-nearest neighbors ($K \in \{3, 5\}$) regressors or an MLP. Similar to personalized calibration, we report summary gaze errors (in millimeters) in the horizontal and vertical directions (Table III).

Consistent with earlier observations, the domain-adapted model again exhibited poor performance across all calibration configurations (see Table III). However, we observed that the domain-adapted model exhibited lower error variability across calibration runs, as evidenced by smaller error deviations. To further investigate this behavior, we evaluated the coefficient of variation for each configuration to assess the relative consistency of performance. We observed that the domain-adapted model consistently exhibited the lowest coefficient of variation across all calibration configurations (see Table IV). This finding suggests that, despite its overall poorer performance, the domain-adapted model provides more predictable and stable gaze estimates, with reduced relative variability compared to the other models. Such stability could be advantageous in applications where consistent performance is prioritized over absolute accuracy. Further, this indicates the potential of a domain-adapted model to produce refined estimates using alternative calibration and error correction mechanisms.

V. DISCUSSION

The results show that an end-to-end integration of face detection, gaze estimation, and gaze-to-screen mapping components can provide stable and interpretable attention signals across multiple users. Although the system does not match the precision of dedicated eye trackers, it operates at a different and practically useful point in the design space, enabling scalable, low-cost, and minimally intrusive observation of attention in natural shared environments. Instead of pixel-level gaze localization, the pipeline reliably identifies regions of interest, attention shifts, and coordinated viewing behavior

Model	3-NN (5-point)		5-NN (9-point)		MLP (9-point)	
	Horz.	Vert.	Horz.	Vert.	Horz.	Vert.
Baseline model	298 ± 97	212 ± 22	268 ± 136	202 ± 34	287 ± 111	219 ± 31
$\lambda_r = 0.1$ (Source only)	298 ± 102	213 ± 18	275 ± 122	208 ± 39	290 ± 105	215 ± 29
$\lambda_r = 0.1, \lambda_c = 0.0$ (Source + Target)	294 ± 97	216 ± 22	259 ± 129	202 ± 25	271 ± 106	214 ± 33
$\lambda_r = 0.1, \lambda_c = 0.1$ (Source + Target)	340 ± 92	223 ± 24	319 ± 112	219 ± 34	320 ± 93	224 ± 25

TABLE I

GAZE ERROR (IN MILLIMETERS) FOR HORIZONTAL AND VERTICAL DIRECTIONS USING 3-NEAREST NEIGHBOR (3-NN), 5-NEAREST NEIGHBOR (5-NN), AND MULTI-LAYER PERCEPTRON (MLP) CALIBRATION MODELS WITH FIVE-POINT AND NINE-POINT CALIBRATION.

Model	3-NN		5-NN		MLP	
	Horz.	Vert.	Horz.	Vert.	Horz.	Vert.
Baseline	0.325	0.101	0.507	0.167	0.387	0.139
$\lambda_r = 0.1$ (Source only)	0.345	0.082	0.442	0.186	0.362	0.135
$\lambda_r = 0.1, \lambda_c = 0.0$ (Source + Target)	0.330	0.102	0.499	0.165	0.372	0.150
$\lambda_r = 0.1, \lambda_c = 0.1$ (Source + Target)	0.271	0.108	0.350	0.156	0.291	0.110

TABLE II

COEFFICIENT OF VARIATION OF GAZE ERROR FOR 3-NN, 5-NN, AND MLP CALIBRATION MODELS WITH 5-POINT AND 9-POINT CALIBRATION.

Model	3-NN		5-NN		MLP	
	Horz.	Vert.	Horz.	Vert.	Horz.	Vert.
Baseline model	279 ± 109	216 ± 30	275 ± 101	209 ± 28	327 ± 85	236 ± 29
$\lambda_r = 0.1$ (Source only)	302 ± 105	235 ± 33	296 ± 98	226 ± 30	308 ± 67	227 ± 18
$\lambda_r = 0.1, \lambda_c = 0.0$ (Source + Target)	277 ± 109	227 ± 35	270 ± 100	218 ± 30	295 ± 69	231 ± 24
$\lambda_r = 0.1, \lambda_c = 0.1$ (Source + Target)	338 ± 79	238 ± 21	329 ± 69	230 ± 17	337 ± 51	237 ± 4

TABLE III

GAZE ERROR (IN MILLIMETERS) FOR HORIZONTAL AND VERTICAL DIRECTIONS USING 3-NEAREST NEIGHBOR (3-NN), 5-NEAREST NEIGHBOR (5-NN), AND MULTI-LAYER PERCEPTRON (MLP) WHEN CALIBRATED BASED ON SESSIONS.

Model	3-NN		5-NN		MLP	
	Horz.	Vert.	Horz.	Vert.	Horz.	Vert.
Baseline	0.393	0.143	0.369	0.137	0.260	0.122
$\lambda_r = 0.1$ (Source only)	0.349	0.139	0.333	0.135	0.218	0.078
$\lambda_r = 0.1, \lambda_c = 0.0$ (Source + Target)	0.396	0.153	0.372	0.140	0.235	0.102
$\lambda_r = 0.1, \lambda_c = 0.1$ (Source + Target)	0.232	0.087	0.211	0.074	0.150	0.061

TABLE IV

COEFFICIENT OF VARIATION OF GAZE ERROR FOR 3-NN, 5-NN, AND MLP WHEN CALIBRATED BASED ON SESSIONS.

between participants, which are often the primary requirements in collaborative interaction analysis.

The approach highlights an important trade-off between accuracy and accessibility. Commodity cameras operate at lower sampling rates and cannot capture subtle eye movement phenomena such as microsaccades, making the system unsuitable for applications requiring clinical or physiological measurement precision. However, many real-world interaction studies depend mainly on spatial attention rather than detailed oculomotor dynamics. For such use cases, coarse gaze localization is sufficient and significantly reduces setup complexity, calibration burden, and equipment cost. The collected dataset further supports this observation, demonstrating that multi-user attention patterns can be studied without equipping each participant with a dedicated device.

The broader impact of this work lies in democratizing gaze-based behavioral analysis. Traditional eye-tracking research depends on expensive specialized hardware, limiting participation to well-funded laboratories and controlled environments. By enabling multi-user gaze analysis using widely available webcams, the proposed approach lowers the barrier to entry for educational institutions, smaller research groups, and public-

sector organizations. This opens opportunities to study collaboration, engagement, and group interaction in classrooms, libraries, museums, and workplaces at a scale previously impractical. In particular, understanding shared attention can inform the design of public displays, collaborative software, and learning technologies that adapt to collective user behavior rather than individual interaction alone.

At the same time, wider accessibility introduces ethical considerations. Because remote gaze estimation can be performed unobtrusively, careless deployment could lead to passive monitoring of individuals without adequate awareness. The technology therefore must be applied with transparency, informed consent, and clear data governance practices. Emphasis should be placed on aggregate behavioral insights rather than individual tracking, and systems should avoid identity inference whenever possible. When used responsibly, the technology can support assistive and adaptive environments rather than surveillance-oriented applications.

Overall, this work positions commodity multi-user gaze tracking as a practical tool for understanding user attention in everyday settings. Rather than replacing high-precision eye trackers, it complements them by enabling large-scale,

ecologically valid studies that were previously difficult to conduct, while also highlighting the importance of ethical deployment alongside technical feasibility.

VI. CONCLUSION

We evaluated the feasibility of multi-user gaze tracking for coarse-grained attention analysis using commodity web cameras and existing off-the-shelf models. Rather than proposing a new framework, our work demonstrates how an end-to-end integration of prior gaze estimation, face detection, and gaze-to-screen mapping methods can support dyadic interactions on large displays. To facilitate this evaluation, we collected a multi-user gaze dataset capturing joint and non-joint attention behaviors. End-to-end experiments show that standard gaze estimation models, combined with simple mapping techniques, can provide consistent coarse-grained gaze localization for multiple users simultaneously, without requiring specialized hardware. These results highlight the potential of leveraging low-cost webcams and existing models to study attention and interaction in shared spaces, providing a practical alternative to traditional, high-cost eye tracking setups. This work lays the foundation for future exploration of scalable multi-user gaze analytics in real-world collaborative environments.

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